CENTRALIZED MATCHING BANDITS: LOWER BOUNDS, TWO-SIDED LEARNING, AND ROBUSTNESS TO ADVERSARIAL CORRUPTION

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ABSTRACT

We study three extensions of the centralized matching-market bandit model of Liu et al. (2020); the original experiments are reproduced in Appendix A. **(i)** For the two-agent two-arm instance in Example 2, we prove a Lai–Robbins-style lower bound showing that the regret of the lower-priority player has leading logarithmic constant $2\gamma/\Delta^2$, which confirms that the dependence on the reward gap is intrinsic to the instance rather than an artefact of the analysis. On the same instance, we also derive the exact ETC regret; its leading constant is a factor of two above the lower bound. **(ii)** We study the two-sided variant in which the arm-side preferences are also unknown and must be learned. Using the binary stable regret as the metric, we show empirically that both UCB and ETC transition from sublinear-looking to near-linear regret as the arm-preference gap shrinks, which identifies arm-side uncertainty as the dominant source of instability. **(iii)** We add a per-player reward corruption step to the protocol, with three adversary classes of increasing power, and propose GS-UCB-M, a Gale–Shapley UCB algorithm whose per-arm estimator is a chronological median of means with a $\sqrt{k}$-inflated confidence radius. The construction of GS-UCB-M does not use the corruption budget C as an input, and the standard sample-mean UCB-GS is recovered as the $k \equiv 1$ special case. Across the three adversary classes, GS-UCB-M converts a bounded corruption budget into a bounded regret plateau and outperforms the sample-mean baseline under the weak and the strong adversaries. Our code is available at inori0cc/rldmuu_seminar_matching_and_bandit.

1 INTRODUCTION

Two-sided online platforms such as labor markets, school choice, online dating and recommender systems — must simultaneously learn agent preferences from noisy feedback and produce a matching that respects two-sided stability. The seminal work of Liu et al. (2020) formalised this problem as a multi-agent bandit on a Gale–Shapley platform and showed that running upper-confidence-bound (UCB) estimators on top of the deferred-acceptance procedure achieves $\mathcal{O}(\log n)$ agent-pessimal stable regret in the centralized setting.

The model of Liu et al. (2020) rests on three strong assumptions. First, the upper bound on the achievable regret leaves open whether the small-gap instances are intrinsically hard or whether the dependence on the reward gap is an artefact of the analysis. Second, the arm-side preferences are assumed to be known to the platform, which fails in markets where the arms (such as employers or recommendation slots) cannot specify their utilities in advance. Third, the reward stream is assumed to be purely stochastic, which ignores the possibility that adversarial actors may inject noise into the observations to bias the matching in their favor. After first reproducing the original empirical results, we tighten each of these three assumptions in turn.

Contributions. The contributions of this paper are organised into three extensions of the original framework.

- **Extension 1 — Lower bound for Example 2** (Section 3). For the two-agent two-arm instance we prove the bound $\liminf_{n\to\infty} R_2^A(n)/\log n \geq 2\gamma/\Delta^2$ for any globally uniformly

good algorithm A. The proof reduces the matching market to a single-player Gaussian bandit and applies the Lai–Robbins lower bound (Lai & Robbins, 1985). We also compute the exact ETC regret constant on the same instance.

- **Extension 2 — Unknown arm preferences** (Section 4). We relax the assumption that arm-side rankings are known and let both sides learn from bandit feedback. Using the binary stable regret as the metric, we compare three algorithms on a controlled 3×3 market and demonstrate a sharp transition driven by the arm-preference gap.
- **Extension 3 — Robust GS-UCB under adversarial corruption** (Section 5). We instantiate the three-tier adversary hierarchy of Wang et al. (2025) with per-player budgets and propose GS-UCB-M, a Gale–Shapley UCB algorithm that uses a chronological median of means with a $\sqrt{k}$-inflated confidence radius. We then show empirically that GS-UCB-M is robust to the three adversary classes without taking the corruption budget as an input.

2 SETTING OF ORIGINAL PAPER

2.1 CENTRALIZED MATCHING MARKET WITH BANDIT FEEDBACK

We follow the centralized matching-bandit setting of Liu et al. (2020). There are N players $[N] = \{1, \dots, N\}$ and $K \geq N$ arms $[K] = \{1, \dots, K\}$. For each player–arm pair (i, j), player i receives a stochastic reward from arm j with unknown mean $\mu_{i,j}$. In the replication experiments, rewards are Gaussian with variance 1, matching the experimental setup of the original paper; we do not clip rewards during replication.

Each arm j has a strict preference ordering π_j over players. In the centralized setting considered here, these arm-side rankings are known to the platform. The players, however, do not know their reward means $\mu_{i,j}$ and must learn their induced preferences over arms from bandit feedback.

At each round $t \in [n]$, the interaction is:

1. Each player i submits a strict ranking $\sigma_i(t)$ over the K arms. In UCB-GS, this ranking is induced by the player's current upper confidence bounds.
2. The platform runs agent-proposing Gale–Shapley (Gale & Shapley, 1962) on the submitted player rankings $\{\sigma_i(t)\}_i$ and the known arm rankings $\{\pi_j\}_j$, producing a stable matching m_t.
3. Each player i observes only the reward from its matched arm, $X_{i,m_t(i)}(t)$, and updates its local statistics for that arm.

Thus the learning problem is not a collection of independent bandits. The platform couples the players through stable matching: a player's ranking can affect not only her own match, but also the matches and future observations of other players.

Stable matching benchmarks. Given the true player preferences induced by the means $\mu_{i,j}$ and the true arm rankings $\{\pi_j\}$, there can be multiple stable matchings. We write $\bar{m}$ for the agent-optimal stable matching, i.e., the stable matching that gives each player the best arm she can obtain in any stable matching. We write $\underline{m}$ for the agent-pessimal stable matching, i.e., the least favorable stable matching for the players.

The corresponding per-player regrets are

$$\bar{R}_i(n) = n\,\mu_{i,\bar{m}(i)} - \sum_{t=1}^{n} \mu_{i,m_t(i)}, \qquad \underline{R}_i(n) = n\,\mu_{i,\underline{m}(i)} - \sum_{t=1}^{n} \mu_{i,m_t(i)}. \tag{1}$$

The distinction between $\bar{R}_i$ and $\underline{R}_i$ is central to the paper. Centralized UCB-GS can have logarithmic regret relative to the agent-pessimal stable benchmark, but the agent-optimal benchmark is stronger and can be impossible to achieve sublinearly in some markets.

2.2 THE UCB-GS ALGORITHM

We reproduce the centralized UCB-GS algorithm. Each player maintains, for every arm j, a successful-pull count $T_{i,j}(t)$ and an empirical mean $\widehat{\mu}_{i,j}(t)$ based only on the rewards observed when player i was actually matched to arm j. Before submitting the ranking at round t, player i computes

$$U_{i,j}(t) = \begin{cases} +\infty, & T_{i,j}(t-1) = 0, \\ \widehat{\mu}_{i,j}(t-1) + \sqrt{\dfrac{3\log t}{2T_{i,j}(t-1)}}, & T_{i,j}(t-1) > 0. \end{cases} \tag{2}$$

The submitted ranking $\sigma_i(t)$ orders arms by decreasing $U_{i,j}(t)$, with deterministic tie-breaking by arm index in our implementation. The platform then runs Gale–Shapley using these submitted rankings. Untried arms receive score $+\infty$, ensuring that each player continues to explore arms it has not yet successfully pulled.

This procedure combines two mechanisms: UCB handles statistical exploration of unknown player-side rewards, while Gale–Shapley enforces a stable matching with respect to the rankings submitted at that round.

3 EXTENSION 1: LOWER BOUND FOR EXAMPLE 2

3.1 MOTIVATION AND INSTANCE

The empirical curve in Figure 4a suggests that player 2's regret grows like $1/\Delta^2$ in Example 2 of Liu et al. (2020). This raises a natural question: is the dependence on player 1's small reward gap merely an artefact of the UCB-GS analysis, or is it intrinsic to the market instance itself? We show that it is intrinsic. The key mechanism is an *externality*: whenever the high-priority player must explore the wrong arm, the low-priority player is displaced from her stable partner and pays regret.

There are two agents and two arms, $\{p_1, p_2\}$ and $\{a_1, a_2\}$. Both arms prefer p_1:

$$a_1 : \ p_1 \succ p_2, \qquad a_2 : \ p_1 \succ p_2.$$

For fixed $\Delta > 0$ and $\gamma > 0$, rewards are Gaussian with variance one:

$$X_{1,a_1} \sim \mathcal{N}(\Delta, 1), \quad X_{1,a_2} \sim \mathcal{N}(0, 1), \qquad X_{2,a_1} \sim \mathcal{N}(0, 1), \quad X_{2,a_2} \sim \mathcal{N}(\gamma, 1).$$

Thus p_1 prefers a_1 while p_2 prefers a_2. The agent-optimal stable matching is

$$m^\star(p_1) = a_1, \qquad m^\star(p_2) = a_2.$$

For an algorithm A, let $R_i^A(n;\theta)$ denote the agent-optimal stable regret of agent p_i at horizon n. We say that A is globally uniformly good on a class $\mathcal{C}$ if, for every instance $\theta \in \mathcal{C}$, every agent p_i, and every $\alpha > 0$, $R_i^A(n;\theta) = o(n^\alpha)$.

3.2 MAIN THEOREM

Let

$$T_{1,2}^A(n) := \sum_{t=1}^{n} \mathbf{1}\{p_1 \text{ is matched to } a_2 \text{ at time } t\}.$$

The key observation is that, because both arms rank p_1 above p_2, the accepted-arm process of p_1 is exactly a two-armed Gaussian bandit process. Whenever the market algorithm matches p_1 to a_j, we can view this as a single-agent algorithm pulling arm a_j and observing p_1's reward. This coupling turns the matching-market externality into a standard bandit identification problem for p_1.

Theorem 1 (Two-by-two externality lower bound)**.** *Let $\mathcal{C}$ contain the Gaussian two-by-two embeddings described above. If A is globally uniformly good on $\mathcal{C}$, then on the instance $\theta_{\Delta,\gamma}$,*

$$\liminf_{n\to\infty} \frac{R_2^A(n;\theta_{\Delta,\gamma})}{\log n} \geq \frac{2\gamma}{\Delta^2}. \tag{3}$$

Proof sketch. The induced single-agent problem for p_1 has arms $\mathcal{N}(\Delta, 1)$ and $\mathcal{N}(0, 1)$. The Lai–Robbins lower bound therefore implies that any globally uniformly good algorithm must pull the suboptimal arm a_2 at least

$$\left(\frac{2}{\Delta^2} + o(1)\right) \log n$$

times. Each such pull forces p_2 away from her stable partner a_2 and onto a_1, causing exactly γ regret in the two-by-two market. Multiplying the two terms gives the leading constant $2\gamma/\Delta^2$. The full coupling argument, the sub-Gaussian interpretation, and the exact ETC derivation are provided in Appendix B.

3.3 ETC COMPARISON

The lower bound also clarifies the cost of the standard explore-then-commit strategy. For a balanced centralized exploration schedule in which each player obtains h successful samples from each arm, the exact regret of player 2 is

$$R_2^{\mathrm{ETC}}(n) = h\gamma + (n - 2h)\gamma\, \Phi\left(-\Delta\sqrt{\frac{h}{2}}\right), \tag{4}$$

where Φ is the standard normal distribution function. Optimising the right-hand side gives

$$h^\star_{\mathrm{ETC}} = \frac{4}{\Delta^2}\log n + O(\log\log n), \qquad R_2^{\mathrm{ETC}}(n) = \left(\frac{4\gamma}{\Delta^2} + o(1)\right)\log n.$$

Thus ETC matches the qualitative $1/\Delta^2$ dependence but loses a factor of two in the leading logarithmic constant relative to the information-theoretic lower bound in Theorem 1. This is the main theoretical message of Extension 1: the spillover effect in Figure 4a is unavoidable, but the usual ETC schedule is not asymptotically sharp on this instance.

4 EXTENSION 2: UNKNOWN ARM PREFERENCES

4.1 MOTIVATION

In the model of Section 2, arm-side preferences $\{\pi_j\}$ are assumed known. Yet in many practical settings — labor markets where employers must screen candidates from noisy interviews, recommendation slots whose CTRs are themselves random variables — the arm side also faces a learning problem. We relax this assumption and study the resulting *two-sided learning* regime.

4.2 PROBLEM SETTING

Both sides have hidden mean reward matrices. Player i's mean for arm j is $\mu_i(j)$ as before; symmetrically, arm j's mean for player i is $\nu_j(i)$. Both sides receive bandit feedback from their match and learn over time. The platform now runs Gale–Shapley using *both* sides' learned rankings. Hence a matching may be stable with respect to the estimated rankings used by Gale–Shapley, but unstable with respect to the true preferences.

Since reward regret alone does not indicate whether the produced matching is stable, we use the *binary stable regret*, following the stable-regret viewpoint of Wu et al. (2026):

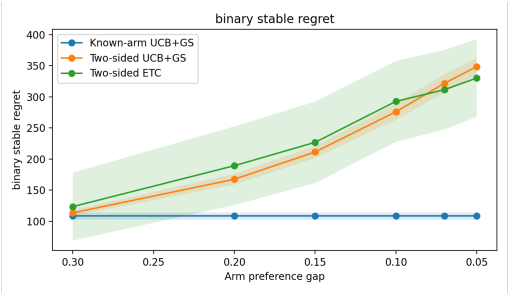
$$\mathrm{BSR}(T) \;=\; \sum_{t=1}^{T} \mathbf{1}\{\, m_t \text{ is unstable}\,\}, \tag{5}$$

where m_t is unstable iff at least one blocking pair exists with respect to *both sides'* true preferences.

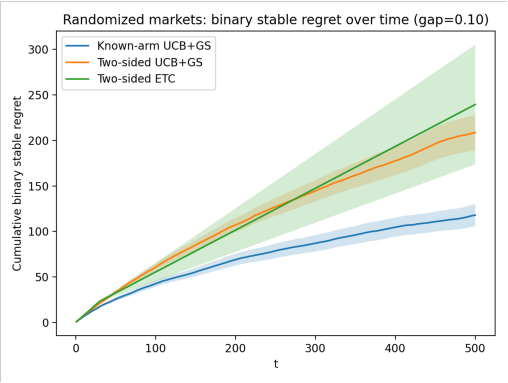
4.3 METHODOLOGY

We compare three algorithms:

- **Known-arm UCB+GS** (baseline): players learn μ, while Gale–Shapley receives the true arm rankings, as in Liu et al. (2020).



(a) Controlled arm-gap sweep.



(b) Randomised-market robustness.

Figure 1: Extension 2 results. (a) Final binary stable regret at horizon T as the arm-preference gap δ decreases. (b) Cumulative binary stable regret over time at fixed $\delta = 0.10$, averaged across randomised markets.

- **Two-sided UCB+GS**: both players and arms maintain UCB estimates; Gale–Shapley uses learned rankings on both sides.
- **Two-sided ETC**: both sides first explore, estimate their rankings, and then Gale–Shapley commits to one matching.

4.4 Experimental methodology

We use two complementary experiments. First, a controlled 3×3 market isolates the effect of arm-side uncertainty. Player-side preferences are kept clear, while the arm-preference gap δ is swept over

$$\delta \in \{0.30, 0.20, 0.15, 0.10, 0.07, 0.05\}.$$

Large δ means that arms can easily distinguish players; small δ makes arm-side rankings harder to learn.

Second, we run a robustness experiment on randomised preference markets. For each trial, player rankings over arms and arm rankings over players are randomised, then converted into utility values with controlled gaps. We fix $\delta = 0.10$ and report cumulative BSR over time, averaged across randomised markets. The key configuration choices are included here; the implementation follows the same Gaussian/no-clipping convention reported in Appendix A.

4.5 Results

Figure 1 highlights two observations. First, the known-arm UCB+GS baseline is almost flat across the sweep in Figure 1a, while Two-sided UCB+GS and Two-sided ETC both rise steeply as $\delta \to 0.05$, indicating that arm-side uncertainty becomes a major source of instability when arm preferences are hard to distinguish. Second, at $\delta = 0.10$ the time-resolved cumulative BSR in Figure 1b grows more slowly for the known-arm baseline, while Two-sided UCB+GS accumulates instability much faster over the simulated horizon. This isolates the effect of replacing the true arm rankings with learned arm rankings inside the same adaptive UCB+GS framework.

These findings suggest that the centralized stability behavior of Liu et al. (2020) relies strongly on the known-arm-rankings assumption. Once arm rankings must also be learned, stable matching becomes significantly harder, especially when arm-side preference gaps are small.

5 EXTENSION 3: ROBUST GS-UCB UNDER ADVERSARIAL CORRUPTION

5.1 MOTIVATION

Real platforms must also defend against adversarial actors — fake ratings, click farms, or coordinated bots — that inject corruption into the reward stream to bias matchings. We import the three-tier adversary hierarchy of Wang et al. (2025) into the matching-market setting and ask whether a UCB-style algorithm can remain robust to all three adversary classes without knowing the total corruption budget.

5.2 PROBLEM SETTING

We retain the centralized matching market of Section 2 and extend the per-round protocol with a corruption step. For Extension 3 the rewards are bounded in $[0, 1]$: the clean stochastic reward of player i from arm j at round t, denoted $r^S_{i,j}(t)$, is drawn i.i.d. from a clipped Gaussian with mean $\mu_{i,j} \in [0, 1]$ and noise standard deviation σ. Three quantities describe the reward stream at each round. The clean matrix $r^S(t)$ contains the values sampled before any adversary acts. The (possibly corrupted) reward matrix $r(t)$ is what the platform passes to Gale–Shapley after any pre-matching adversary has modified it; in the clean case $r(t) = r^S(t)$. The scalar $X_i(t)$ is the observation that the learner of player i consumes after the matching m_t has been produced; in the clean case $X_i(t) = r^S_{i,m_t(i)}(t)$, while a post-matching adversary may instead set $X_i(t)$ to a modified value.

Each adversary class holds an independent per-player budget $C^{\mathrm{rem}}_i(t)$, initialised at C and decreased monotonically; once a budget is exhausted, the adversary becomes a no-op for that player. At round t the adversary picks a target arm $j^\dagger_i(t)$ for player i and applies a single-coordinate boost of magnitude

$$\delta_{i,t} \;=\; \min\big(\varepsilon,\; 1 - r^S_{i,j^\dagger_i(t)}(t),\; C^{\mathrm{rem}}_i(t)\big). \tag{6}$$

The three terms inside the minimum encode three separate constraints on the adversary. The constant $\varepsilon \in (0, 1)$ is a per-round cap that limits the size of any single perturbation. The headroom term $1 - r^S_{i,j^\dagger_i(t)}(t)$ keeps the corrupted value within the reward support $[0, 1]$. The remaining-budget term $C^{\mathrm{rem}}_i(t)$ ensures that the per-round spend cannot exceed the budget still available to the adversary. All three classes share the static anchor target $j^{\mathrm{sub}}_i = \arg\max_{j \neq m^\star(i)} \mu_{i,j}$, the runner-up arm in the true player preference, with ties broken by ascending arm index. Fixing the target across classes ensures that the observed performance differences isolate *when* and *with what information* the adversary acts, rather than the choice of target arm. The two budget accounting conventions of Wang et al. (2025) for pre- and post-matching corruption are applied accordingly, with formal definitions deferred to Appendix C.1.

5.3 THREE ADVERSARY CLASSES

Class A is a weak, oblivious adversary with information set $\mathcal{F}^A_t = \sigma(\boldsymbol{\mu}, m^\star, \varepsilon)$. It targets the fixed anchor $j^\dagger_i(t) \equiv j^{\mathrm{sub}}_i$ at every round and adds $\delta_{i,t}$ to that single entry of the reward row before Gale–Shapley is run. The boost reaches the observation $X_i(t)$ only on the rounds in which the matching assigns $m_t(i) = j^{\mathrm{sub}}_i$, so budget spent on any other round is wasted.

Class B is a medium, adaptive-online adversary with information set $\mathcal{F}^B_t = \sigma\big(\boldsymbol{\mu}, m^\star, \varepsilon, \{(m_s, X_s)\}_{s<t}\big)$. It maintains a mirror tracker that counts $N^{\mathrm{mir}}_{i,j}(t)$ and means $\hat{\mu}^{\mathrm{mir}}_{i,j}(t)$ built from the same post-corruption observations the learner sees, and chooses $j^\dagger_i(t)$ dynamically to displace the current mirror leader. Class B does not spend any budget while some arm has $N^{\mathrm{mir}}_{i,j}(t) = 0$, because untried arms dominate the learner ranking through the $+\infty$ sentinel and no corruption of a pulled coordinate can change the order.

Class C is a strong, adaptive-offline adversary with information set $\mathcal{F}^C_t = \sigma\big(\mathcal{F}^B_t, m_t, r^S(t)\big)$. It acts after Gale–Shapley produces m_t and directly edits the scalar observation: $X_i(t) = r^S_{i,m_t(i)}(t) + \delta_{i,t}$ when $m_t(i) = j^{\mathrm{sub}}_i$, and $r^S_{i,m_t(i)}(t)$ otherwise, followed by a clip to $[0, 1]$. Class C is strictly more budget-efficient than Class A because every unit of attack budget translates one-to-one into observed perturbation. The information hierarchy $\mathcal{F}^A \subset \mathcal{F}^B \subset \mathcal{F}^C$ holds by construction.

5.4 THE GS-UCB-M ALGORITHM

A single corrupted sample of size Δ shifts the sample mean by Δ/N; over b corrupted samples the shift is $b\Delta/N$, which grows without bound as the adversary spends budget and is worst when N is small. We therefore replace the sample mean by a chronological median of means. The N samples for arm j are partitioned into k blocks, the block means $G_1, \ldots, G_k$ are computed, and the reported estimate is their median. Corruption confined to fewer than $\lfloor k/2 \rfloor$ blocks is rejected. The chronological partition is essential: a budget concentrated in a short time window touches only a few consecutive samples, which a chronological partition keeps inside neighbouring blocks; a random partition would spread the contamination across most blocks and defeat the median.

For arm j with $N = N_{i,j}(t)$ chronological observations $(x_1, \ldots, x_N)$, group count $k = k_{i,j}(t)$, and block size $g = \lfloor N/k \rfloor$, the estimator is

$$\hat{\mu}^{\mathrm{m}}_{i,j}(t) \;=\; \mathrm{median}(G_1, \ldots, G_k), \qquad G_\ell \;=\; \frac{1}{g} \sum_{m=(\ell-1)g+1}^{\ell g} x_m. \tag{7}$$

We use the schedule $k_{i,j}(t) = \min\big(2\lceil \log T \rceil + 1,\ \lfloor \sqrt{N} \rfloor\big)$, kept odd so that the median is unambiguous. The upper cap is the standard $k = \Theta(\log T)$ choice from the median-of-means literature, and the lower cap $\sqrt{N}$ is a practical safeguard that keeps each block large enough to average out within-block noise. The two caps together trade robustness (more blocks tolerate more corrupted samples) against statistical efficiency (fewer blocks give lower variance per block mean). Because each block mean averages only N/k samples, the standard deviation of $\hat{\mu}^{\mathrm{m}}_{i,j}$ is a factor of $\sqrt{k}$ larger than that of the sample mean, so we inflate the UCB radius by $\sqrt{k}$:

$$W_{i,j}(t) \;=\; \sqrt{\frac{\alpha_{\mathrm{m}} \cdot k_{i,j}(t) \cdot \log\big(\max(t,2)\big)}{\max(N_{i,j}(t),\, 1)}}, \tag{8}$$

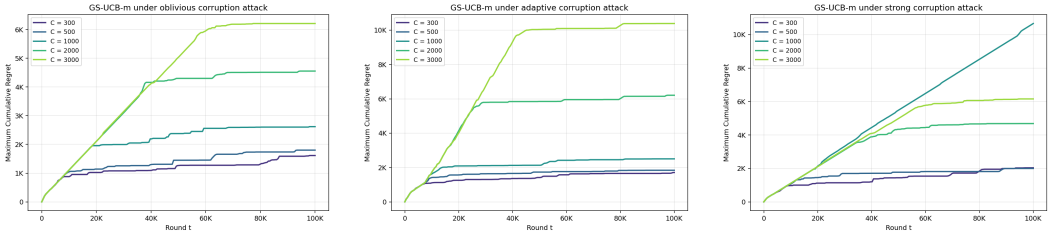
where $\alpha_{\mathrm{m}} > 0$ is a fixed UCB exploration coefficient (set to 0.05 throughout our experiments) and the convention $\log(\max(t,2))$ avoids the boundary case $\log 1 = 0$ at $t = 1$. The player-side index is $U_{i,j}(t) = \hat{\mu}^{\mathrm{m}}_{i,j}(t) + W_{i,j}(t)$, with $U_{i,j}(t) = +\infty$ when $N_{i,j}(t) = 0$. By construction, neither the schedule of $k_{i,j}(t)$ nor the radius $W_{i,j}(t)$ involves the corruption budget C, so the algorithm is C-agnostic; setting $k \equiv 1$ collapses the median of means to the sample mean and the $\sqrt{k}$ factor to one, recovering the sample-mean UCB-GS of Liu et al. (2020) as a strict special case.

At small N, the median of k block means is statistically unstable, so we apply the median of means only once enough samples have accumulated. Concretely, an *activation threshold* N_{act} controls the switch: while $N_{i,j}(t) < N_{\mathrm{act}}$, the estimator $\hat{\mu}_{i,j}(t)$ falls back to the plain sample mean, and once $N_{i,j}(t) \geq N_{\mathrm{act}}$ the chronological median of means in (7) takes over. The full pseudocode of GS-UCB-M, together with the threshold N_{act}, the caching scheme, and the determinism guarantees, is given in Algorithm 1 of Appendix C.3.

5.5 EXPERIMENTS

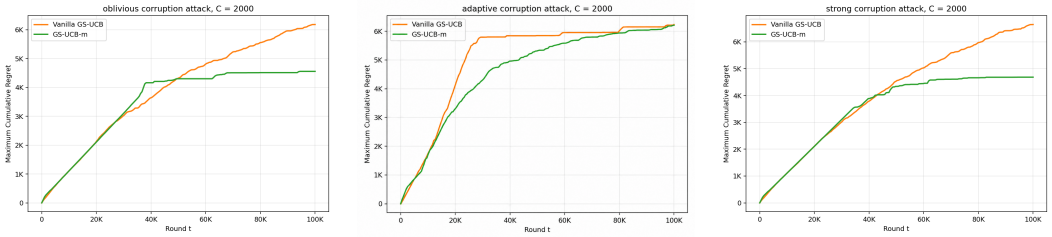
The market has $N = K = 5$ with mean-reward gap $\Delta = 0.1$ built from per-player rank permutations; the horizon is $T = 100{,}000$; and the corruption budgets are $C \in \{300,\ 500,\ 1000,\ 2000,\ 3000\}$. The baseline is Vanilla UCB-GS (Liu et al., 2020); the primary metric is the clean stable pseudo-regret against $m^\star$, aggregated as $\mathrm{MaxReg}(t) = \max_i R_i(t)$, and the reported curves are averaged over 3 seeds. The full experimental protocol is given in Appendix C.6.

Result 1: GS-UCB-M absorbs corruption across all three adversaries. Figure 2 shows three regularities. Across all five budgets and all three adversaries, the regret curve plateaus once C is exhausted, so the algorithm converts a bounded attack budget into bounded total regret. The plateau height grows monotonically with C, and small budgets ($C \leq 1000$) cause negligible long-run damage. Class B yields the sharpest plateau because it does not act until every arm has been pulled at least once, so the entire budget is spent in a short interval. The chronological median of means is most effective when the corrupted samples lie in only a few consecutive blocks, which is precisely this case.



(a) Class A — weak / oblivious. (b) Class B — medium / adaptive. (c) Class C — strong / post-matching.

Figure 2: MaxReg of GS-UCB-M under each adversary class for budgets $C \in \{300, 500, 1000, 2000, 3000\}$. The regret plateaus once the adversary has exhausted C.



(a) Class A, $C = 2000$. (b) Class B, $C = 3000$. (c) Class C, $C = 2000$.

Figure 3: MaxReg of Vanilla UCB-GS and GS-UCB-M at a fixed budget, for each adversary class.

Result 2: Comparison with Vanilla UCB-GS. Figure 3 compares the two algorithms at fixed budgets. Under Class A and Class C, the regret of Vanilla UCB-GS grows roughly linearly once the adversary begins to act, while GS-UCB-M saturates and stays bounded; GS-UCB-M therefore performs better than the baseline in these two settings. Under Class B, GS-UCB-M still saturates, but its plateau is close to or slightly worse than the baseline. Because Class B can read the mirror tracker, it spreads its corruption across many rounds rather than concentrating it; the corrupted samples then fall into many different blocks instead of a few neighbouring ones, and fewer of them are rejected by the median. The chronological median of means is therefore most effective when the corruption is concentrated in a short interval, and adapting the block partition to the spending pattern of the adversary is a natural direction for future work.

6 Conclusion

We revisited the centralized matching-market bandit model of Liu et al. (2020) and extended it along three axes. Extension 1 is the main theoretical contribution. For Example 2 we proved a $2\gamma/\Delta^2$ lower bound on the leading logarithmic constant of the regret of the lower-priority player, and we derived the exact ETC leading constant $4\gamma/\Delta^2$ on the same instance, which is a factor of two above the lower bound. Extension 2 showed empirically that, once the arm-side preferences must also be learned, stable matching becomes a substantially harder two-sided learning problem. Extension 3 showed that the chronological median of means improves the robustness of centralized UCB-GS to bounded reward corruption, with the largest gain observed under the weak and the strong adversaries.

The main limitation of this work is that Extensions 2 and 3 are purely empirical. Formal stable-regret bounds for the two-sided learning regime and for the chronological median-of-means estimator under adversarial corruption remain natural directions for future work.

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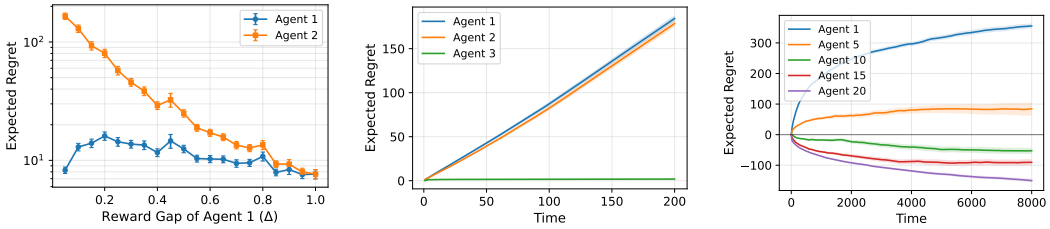
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A Replication

A.1 Replication results

We re-implemented centralized UCB-GS from scratch and reproduced the three experiments shown in Figure 1 of Liu et al. (2020). The example numbering follows the original paper: the three panels correspond to Example 2, Example 6, and Example 7. The numbering is not consecutive because these are the examples selected by the authors for Figure 1, while the omitted examples serve other theoretical or explanatory roles.



(a) Example 2: small-gap spillover. (b) Example 6: UCB lock-in. (c) Example 7: global preferences.

Figure 4: Reproduction of Figure 1 from Liu et al. (2020). (a) In the two-agent two-arm market of Example 2, player 2's regret increases as player 1's reward gap becomes small. (b) In Example 6, agents 1 and 2 incur near-linear agent-optimal regret, illustrating that UCB-GS is not generally sublinear under the agent-optimal benchmark. (c) In the global-preference market of Example 7, lower-ranked agents can obtain negative stable regret because exploration sometimes gives them arms better than their stable benchmark.

Figure 4 matches the qualitative phenomena reported in the original paper:

- Panel 4a shows the small-gap spillover effect: player 2's regret grows when player 1 cannot quickly identify her preferred arm. This motivates the lower-bound analysis in Section 3.
- Panel 4b shows that the agent-optimal regret of centralized UCB-GS can be near-linear, even though the agent-pessimal benchmark admits logarithmic guarantees.
- Panel 4c reproduces the negative-regret phenomenon for low-ranked agents in markets with global preferences.

A.2 Implementation

Each sub-figure of Figure 4 uses the per-experiment settings:

- Figure 4a: $N = K = 2$, $n = 400$, 100 trials per $\Delta \in \{0.05, 0.10, \ldots, 1.00\}$ (20-point uniform grid); base seed `20240101`, offset by the Δ-index across the sweep.
- Figure 4b: $N = K = 3$, $n = 200$, 100 trials; master seed `20240102`. Example 6 parameter `p3_gap` $= 0.05$ (the a_3–a_1 mean gap for agent p_3).

- Figure 4c: $N = K = 20$, $n = 8000$, 50 trials; master seed `20240103`. Example 7 uses global preferences with consecutive-arm mean gap $\texttt{gap} = 0.1$ (so $\mu_{i,j} = (N-1-j) \cdot 0.1$).

All experiments use Gaussian rewards with unit standard deviation ($\sigma = 1$, no clipping), matching the original paper. We implemented agent-proposing Gale–Shapley from scratch with $\mathcal{O}(NK)$ work per call. The UCB exploration constant matches Equation (2); arms with $T_{i,j} = 0$ score $+\infty$ and are placed at the head of the ranking in uniformly random order. A test suite of 80 unit tests passes against the implementation (31 explicit tests + a 50-seed parametrized GS stability sweep).

B Extension 1: Proofs and ETC Derivations

B.1 Proof of Theorem 1

We first formalise the coupling used in the main text. Construct a single-agent bandit algorithm $A^{\#}$ from the matching-market algorithm A. The algorithm $A^{\#}$ internally simulates the two-agent market. Whenever the simulated market matches p_1 to a_j, $A^{\#}$ pulls arm a_j in the single-agent Gaussian bandit and feeds the observed reward back into the simulated history of p_1. The rewards of p_2 can be generated internally because their distributions are fixed and do not affect which arm p_1 can obtain: both arms rank p_1 above p_2.

Under this coupling,

$$T_2^{A^{\#}}(n) = T_{1,2}^{A}(n),$$

where $T_2^{A^{\#}}(n)$ is the number of pulls of the second arm by $A^{\#}$ and $T_{1,2}^{A}(n)$ is the number of times p_1 is matched to a_2 in the market. The global-uniform-good condition ensures that $A^{\#}$ is uniformly good over the corresponding Gaussian alternatives. Hence the Lai–Robbins lower bound gives

$$\liminf_{n\to\infty} \frac{\mathbb{E}[T_2^{A^{\#}}(n)]}{\log n} \geq \frac{1}{\mathrm{KL}(\mathcal{N}(0,1) \,\|\, \mathcal{N}(\Delta,1))}.$$

For unit-variance Gaussians,

$$\mathrm{KL}(\mathcal{N}(0,1) \,\|\, \mathcal{N}(\Delta,1)) = \frac{\Delta^2}{2},$$

and therefore

$$\liminf_{n\to\infty} \frac{\mathbb{E}_\theta[T_{1,2}^{A}(n)]}{\log n} \geq \frac{2}{\Delta^2}.$$

Finally, in the two-by-two complete market, whenever p_1 is matched to a_2, agent p_2 is matched to a_1. Since p_2's stable benchmark is a_2 with mean γ and a_1 has mean 0, each such round contributes exactly γ regret. Thus

$$R_2^{A}(n; \theta_{\Delta,\gamma}) \geq \gamma \, \mathbb{E}_\theta[T_{1,2}^{A}(n)],$$

which proves Theorem 1.

B.2 Sub-Gaussian interpretation

The lower bound transfers directly to the 1-sub-Gaussian model. Indeed, $\mathcal{N}(\mu, 1)$ is 1-sub-Gaussian, so the Gaussian instance above is a member of the 1-sub-Gaussian class. Hence, if an algorithm is globally uniformly good for all 1-sub-Gaussian matching-bandit instances, then it is also globally uniformly good on the Gaussian subclass, and the same lower bound follows. This is a subclass lower bound for the sub-Gaussian model; it is not a claim that every possible sub-Gaussian family has the same exact information constant.

B.3 Exact ETC regret derivation

Consider centralized ETC with exploration parameter h under the balanced centralized exploration schedule used in Section 3. During the first $2h$ rounds, each player obtains h successful samples from each arm. For p_2, exploration regret is

$$h(\mu_2(a_2) - \mu_2(a_1)) + h(\mu_2(a_2) - \mu_2(a_2)) = h\gamma.$$

In the commit phase, p_2 suffers regret only if p_1's empirical ranking is wrong, i.e. if $\widehat{\mu}_{1,a_2} \geq \widehat{\mu}_{1,a_1}$. Since

$$\widehat{\mu}_{1,a_1} - \widehat{\mu}_{1,a_2} \sim \mathcal{N}\left(\Delta, \frac{2}{h}\right),$$

we have

$$\mathbb{P}(\widehat{\mu}_{1,a_2} \geq \widehat{\mu}_{1,a_1}) = \Phi\left(-\Delta\sqrt{\frac{h}{2}}\right).$$

This gives the exact expression

$$R_2^{\mathrm{ETC}}(n) = h\gamma + (n-2h)\gamma\Phi\left(-\Delta\sqrt{\frac{h}{2}}\right).$$

The corresponding expression for player 1 is

$$R_1^{\mathrm{ETC}}(n) = h\Delta + (n-2h)\Delta\Phi\left(-\Delta\sqrt{\frac{h}{2}}\right).$$

Differentiating the expression for R_2^{ETC} yields the first-order condition

$$1 = 2\Phi\left(-\Delta\sqrt{\frac{h}{2}}\right) + (n-2h)\frac{\Delta}{2\sqrt{2h}}\phi\left(\Delta\sqrt{\frac{h}{2}}\right),$$

where ϕ is the standard normal density. Asymptotically,

$$h^\star_{\mathrm{ETC}} = \frac{4}{\Delta^2}\log n + O(\log\log n),$$

which gives

$$R_2^{\mathrm{ETC}}(n) = \left(\frac{4\gamma}{\Delta^2} + o(1)\right)\log n.$$

This is a factor two above the lower-bound constant in Theorem 1.

C Extension 3: Detailed Protocol, Adversary Specification, and Implementation

The main text gives the compact version of Extension 3. This appendix provides the full per-round protocol, the budget accounting conventions, the detailed specification of each adversary class, the complete pseudocode of GS-UCB-M, and the hyperparameters used in our experiments.

C.1 Budget accounting and adversary specification

Two budget metrics. Budget accounting follows the two conventions of Wang et al. (2025). For the pre-matching adversaries (Class A and Class B), the per-round cost is the L_∞ distance over the reward row,

$$c_i^{\mathrm{corr}}(t) \;=\; \max_{j\in[K]} \left|r_{i,j}(t) - r^S_{i,j}(t)\right| \;=\; \delta_{i,t}. \tag{9}$$

Because the pre-matching adversaries modify at most one coordinate per round, the maximum is attained at that coordinate and reduces to $\delta_{i,t}$. For the post-matching adversary (Class C), the per-round cost is the absolute deviation at the matched coordinate,

$$c_i^{\mathrm{att}}(t) \;=\; \left|X_i(t) - r^S_{i,m_t(i)}(t)\right|. \tag{10}$$

Either quantity is subtracted from C_i^{rem} at the end of the round in which it is incurred.

Anchor target. The static anchor target used by all three classes is

$$j_i^{\mathrm{sub}} \;=\; \arg\max_{j\neq m^\star(i)} \mu_{i,j}, \qquad \text{(ties broken by ascending arm index).} \tag{11}$$

The anchor is the static target of Class A and Class C, and serves as the asymptotic target of Class B once its mirror estimates have converged.

Class A (weak, oblivious). Class A executes the single-coordinate update

$$r_{i,j_i^{\mathrm{sub}}}(t) \;\leftarrow\; r^S_{i,j_i^{\mathrm{sub}}}(t) + \delta_{i,t}, \qquad r_{i,j}(t) \;=\; r^S_{i,j}(t) \quad \forall\, j \neq j_i^{\mathrm{sub}},$$

before Gale–Shapley is run. The corruption sequence is oblivious in the sense of Wang et al. (2025): any dependence on $r^S(t)$ enters only through the headroom clip in (6), which is a numerical safeguard rather than a strategic adaptation.

Class B (medium, adaptive-online). Class B maintains a per-player mirror tracker:

$$N^{\mathrm{mir}}_{i,j}(t) \;=\; \big|\{s < t : m_s(i) = j\}\big|, \qquad S^{\mathrm{mir}}_{i,j}(t) \;=\; \sum_{s<t:m_s(i)=j} X_i(s),$$

with the derived mean $\hat{\mu}^{\mathrm{mir}}_{i,j}(t) = S^{\mathrm{mir}}_{i,j}(t)/\max\big(N^{\mathrm{mir}}_{i,j}(t), 1\big)$. The mirror is fed the post-corruption observations $X_i(s)$, so it is self-poisoned and stays aligned with the sufficient statistics that the learner maintains from the same stream. The target at round t is

$$j_i^{\dagger}(t) \;=\; \begin{cases} j_i^{\mathrm{top}}(t) & \text{if } j_i^{\mathrm{top}}(t) \neq m^{\star}(i), \\ \arg\max_{j \neq m^{\star}(i)} \hat{\mu}^{\mathrm{mir}}_{i,j}(t) & \text{otherwise,} \end{cases} \qquad j_i^{\mathrm{top}}(t) \;=\; \arg\max_j \hat{\mu}^{\mathrm{mir}}_{i,j}(t), \quad (12)$$

with ascending arm-index tie-breaking throughout. Class B sets $\delta_{i,t} = 0$ whenever any arm of player i has $N^{\mathrm{mir}}_{i,j}(t) = 0$, since the $+\infty$ sentinel on unpulled arms dominates the player ranking and no perturbation of a pulled coordinate can change the order. The mirror is updated end-of-round with $(m_t(i), X_i(t))$, so the quantities used inside (12) depend strictly on $\{X_s\}_{s<t}$.

Class C (strong, adaptive-offline). Class C acts after the matching m_t has been produced and directly edits the scalar observation:

$$X_i(t) \;=\; \begin{cases} r^S_{i,m_t(i)}(t) + \delta_{i,t} & \text{if } m_t(i) = j_i^{\mathrm{sub}}, \\ r^S_{i,m_t(i)}(t) & \text{otherwise,} \end{cases} \quad (13)$$

followed by a final clip of $X_i(t)$ to $[0, 1]$. The target is held fixed at j_i^{sub}, identical to Class A, so the only material difference between the two classes is the timing of the action.

C.2 Activation threshold for the median of means

At small N, the median of k block means is statistically unstable, both because each block mean averages only $g = N/k$ samples and because the median over k values is itself a discrete order statistic. Empirically, this early-round noise can drive the multi-player Gale–Shapley game into a non-canonical stable matching that subsequently locks in for the remainder of the horizon. To mitigate this, we use an activation threshold $N_{\mathrm{act}} = 400$ and substitute the sample mean for the median of means while $N_{i,j}(t) < N_{\mathrm{act}}$ or $k_{i,j}(t) < 3$:

$$\hat{\mu}_{i,j}(t) \;=\; \begin{cases} \frac{1}{N}\sum_{m=1}^{N} x_m & \text{if } N_{i,j}(t) < N_{\mathrm{act}} \text{ or } k_{i,j}(t) < 3, \\ \hat{\mu}^{\mathrm{m}}_{i,j}(t) & \text{otherwise.} \end{cases} \quad (14)$$

The value $N_{\mathrm{act}} = 400$ corresponds approximately to the sample count at which the $\sqrt{N}$ cap on $k_{i,j}(t)$ becomes inactive, so that the median is computed over a sufficient number of block means.

C.3 Pseudocode

Algorithm 1 states the local procedure of a single player. The platform-side step is unchanged from the standard agent-proposing Gale–Shapley: collect every player ranking and run deferred acceptance against the fixed arm-side preferences.

C.4 Caching, storage, and determinism

Only the matched arm receives a new sample in each round, so only one median-of-means entry must be recomputed per player per round. When the partition shape (g, k) and the prefix length

Algorithm 1 GS-UCB-M (local procedure for player i).

Require: Horizon T, arms $[K]$, exploration coefficient α_{m}, activation threshold N_{act}.

Initialise the chronological history $H_{i,j} \leftarrow ()$ and the pull count $N_{i,j} \leftarrow 0$ for every arm $j \in [K]$.
for $t = 1, 2, \ldots, T$ **do**
  for $j = 1, \ldots, K$ **do**
    $k_{\mathrm{cap}} \leftarrow 2\lceil \log T \rceil + 1$; $k_{i,j} \leftarrow \min\big(k_{\mathrm{cap}}, \lfloor \sqrt{N_{i,j}} \rfloor\big)$; subtract 1 if $k_{i,j}$ is even; $k_{i,j} \leftarrow \max(k_{i,j}, 1)$.
    if $N_{i,j} = 0$ **then**
      $U_{i,j} \leftarrow +\infty$.
    else
      $U_{i,j} \leftarrow \hat{\mu}_{i,j} + \sqrt{\alpha_{\mathrm{m}}\, k_{i,j}\, \log(\max(t,2)) / \max(N_{i,j}, 1)}$ ▷ $\hat{\mu}$ from (14)
    end if
  end for
  $\sigma_i \leftarrow$ descending sort of $\{U_{i,j}\}_j$ with ties by ascending arm index.
  Submit σ_i; receive matched arm $j^\star = m_t(i)$ and observe $X_i(t)$.
  Append $X_i(t)$ to $H_{i,j^\star}$ and increment $N_{i,j^\star}$.
  Recompute $\hat{\mu}_{i,j^\star}$ from $H_{i,j^\star}$ via (14).
end for

gk are unchanged from the previous re-computation of arm j, the median is bit-identical and the cached value is reused. Each per-arm history $H_{i,j}$ is stored in a doubling buffer so that reads of the valid prefix are zero-copy views, which keeps the median re-computation memory-friendly at large T. Independent random streams for player permutations, arm-side rankings, reward sampling, and adversary randomness are spawned from a single master seed per run, so two runs of GS-UCB-M with the same master seed produce bit-identical matchings.

C.5 Hyperparameters

Table 1 lists the hyperparameters used in all experiments. The corruption budget C is not an input to the algorithm; GS-UCB-M is C-agnostic by construction.

Parameter	Description	Default value
α_{m}	UCB exploration coefficient in (8)	0.05
N_{act}	Activation threshold for the median of means in (14)	400
k_{cap}	Upper cap on the group count	$2\lceil \log T \rceil + 1$
Log convention	Used to avoid $\log 1 = 0$ at $t = 1$	$\log(\max(t, 2))$
Tie-breaking	Ranking ties broken by arm index	ascending
Untried arms	Initial exploration via $+\infty$ sentinel	$U_{i,j} \leftarrow +\infty$
History storage	Per-pair chronological buffer	doubling `numpy` array

Table 1: Hyperparameters of GS-UCB-M.

C.6 Experimental setup

The market has $N = K = 5$ with per-player rank permutations and a mean-reward gap of $\Delta = 0.1$; the horizon is $T = 100{,}000$; the corruption budgets used in Figure 2 are $C \in \{300, 500, 1000, 2000, 3000\}$, and the budgets used in the comparison of Figure 3 are $C \in \{2000\}$. The algorithms compared are Vanilla UCB-GS (Liu et al., 2020) and GS-UCB-M, and the adversaries are Class A, Class B, and Class C from Appendix C.1. The primary metric is the clean stable pseudo-regret against $m^\star$, aggregated as $\mathrm{MaxReg}(t) = \max_i R_i(t)$. All curves are averaged over 3 seeds.